

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Hemalakshmi H - 192571049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Let the following propositional symbols represent the patient's symptoms and diagnoses:

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
B	Patient has Breathlessness
R	Patient has Rash
Flu	Patient has Possible Flu
M	Patient has Possible Measles
P	Patient has Possible Pneumonia

Rules

Rule I: If a patient has Fever AND Cough $\rightarrow$ Possible Flu

$$F \wedge C \rightarrow Flu$$

Rule II: If a patient has Fever AND Rash $\rightarrow$ Possible Measles

$$F \wedge R \rightarrow M$$

Rule III: If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

$$C \wedge B \rightarrow P$$

Facts about Patient A

$$\begin{aligned} F &= True \\ C &= True \\ B &= True \end{aligned}$$

There is no fact stating that Patient A has Rash, so:

$$R = False/Unknown$$

The initial knowledge base is therefore:

$$KB = \{F, C, B, (F \wedge C \rightarrow Flu), (F \wedge R \rightarrow M), (C \wedge B \rightarrow P)\}$$

(b) Forward Chaining

What is Forward Chaining?

Forward Chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied. New facts are added until no more rules can be fired.

Initial Facts

F,C,B

That is:

Fever = True

Cough = True

Breathlessness = True

Step 1 — Apply Rule I

Rule I:

F ∧ C → Flu

We have:

F = True

and

C = True

Therefore, both conditions are satisfied.

Derive:

Flu = True

So:

Patient A → Possible Flu

Step 2 — Apply Rule II

Rule II:

F ∧ R → M

We have:

F = True

But there is no Rash fact:

R ≠ True

Therefore, Rule II cannot be fired.

So:

M cannot be derived

Step 3 — Apply Rule III

Rule III:

C ∧ B → P

We have:

$C = \text{True}$

and

$B = \text{True}$

Therefore, both conditions are satisfied.

Derive:

$P = \text{True}$

So:

Patient A $\rightarrow$ Possible Pneumonia

Forward Chaining Summary

Step	Rule	Conditions	New Fact
Initial	—	Fever, Cough, Breathlessness	F, C, B
1	Rule I	$F \wedge C$	Flu
2	Rule II	$F \wedge R$	Cannot fire — Rash unavailable
3	Rule III	$C \wedge B$	Pneumonia

Final Derived Knowledge

F, C, B, Flu, P

Therefore, the system derives:

Possible Flu

and

Possible Pneumonia

It cannot derive Possible Measles because Rash is not known.

Important: In this rule-based system, "Possible Pneumonia" is a logical conclusion from the given rules, not a real-world medical diagnosis.

(c) Backward Chaining for Pneumonia
Goal

The goal is:

P

where P = Patient A has Possible Pneumonia.

Backward Chaining starts from the goal and works backward to determine what conditions are required to prove it.

Step 1 — Start with the Goal

$$Goal = P$$

Find a rule whose conclusion is P.

Rule III is:

$$C \wedge B \rightarrow P$$

Therefore, to prove P, we must prove:

$$C \wedge B$$

So the goal is reduced into two subgoals:

$$P \\ \downarrow$$

$$C \text{ AND } B$$

Step 2 — Prove Cough

We need:

$$C$$

Check the knowledge base.

Cough is already an initial fact:

$$C = \text{True}$$

Therefore:

C is proved

Step 3 — Prove Breathlessness

We need:

$$B$$

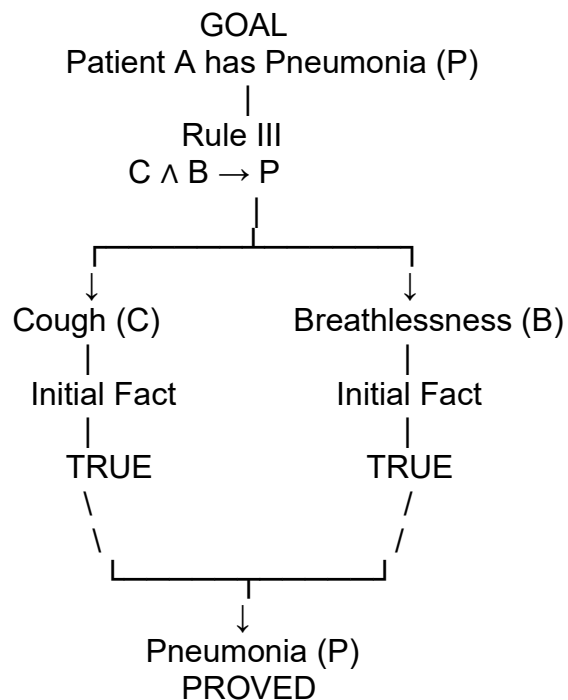
Breathlessness is also an initial fact:

$$B = \text{True}$$

Therefore:

B is proved

Goal Reduction Tree



Backward Chaining Inference

requires: P

Both are facts: $C \wedge B$

Therefore: $C = \text{True}, B = \text{True}$

Hence: $C \wedge B = \text{True}$

$P = \text{True}$

Final Conclusion:

Using Backward Chaining, the goal "Patient A has Possible Pneumonia" is successfully verified because:

$$\text{Cough} \wedge \text{Breathlessness} \rightarrow \text{Pneumonia}$$

and Patient A has both Cough and Breathlessness.

Therefore:

Patient A has Possible Pneumonia

Case Study 2 — Autonomous Traffic Management Agent

This case study uses **First-Order Logic (FOL)** to represent vehicles, emergency situations, traffic signals, and vehicle movement.

(a) Represent the Knowledge Base in FOL

Predicate	Meaning
Vehicle(x)	x is a vehicle
EmergencyType(x)	x is an emergency vehicle/type
ClearPath(x)	A clear path is provided for x
GreenSignal(x)	x has a green traffic signal
RedSignal(x)	x has a red traffic signal
Behind(x,y)	x is behind y
Proceed(x)	x is allowed to proceed

Constants

Constant	Meaning
Ambulance	The emergency ambulance
CarA	A car

Rule 1

If a vehicle is an emergency vehicle, then a clear path is provided for it.

$$\forall x (Vehicle(x) \wedge EmergencyType(x) \rightarrow ClearPath(x))$$

Rule 2

If a vehicle has a clear path, it gets a green signal and does not have a red signal.

$$\forall x (ClearPath(x) \rightarrow GreenSignal(x) \wedge \neg RedSignal(x))$$

This can also be considered as two conclusions:

$$ClearPath(x) \rightarrow GreenSignal(x)$$

and

$$ClearPath(x) \rightarrow \neg RedSignal(x)$$

Rule 3

If vehicle x is behind vehicle y and y has a green signal, then x can proceed.

$$\forall x \forall y (Vehicle(x) \wedge Behind(x,y) \wedge GreenSignal(y) \rightarrow Proceed(x))$$

(b) Given Facts

The initial facts are:

Vehicle(Ambulance)
EmergencyType(Ambulance)
Vehicle(CarA)
Behind(CarA,Ambulance)

So we know:

Ambulance is a vehicle.

Ambulance is an emergency vehicle.

CarA is a vehicle.

CarA is behind the Ambulance.

(c) Apply Forward Chaining

Forward Chaining starts with the known facts and applies rules to derive new facts.

Step 1 — Apply Rule 1

Rule 1:

$Vehicle(x) \wedge EmergencyType(x) \rightarrow ClearPath(x)$

For:

$x = Ambulance$

we have:

$Vehicle(Ambulance) = True$

and

$EmergencyType(Ambulance) = True$

Therefore:

$ClearPath(Ambulance)$

Derived Fact 1

Ambulance has a clear path.

Step 2 — Apply Rule 2

Rule 2:

$ClearPath(x) \rightarrow GreenSignal(x) \wedge \neg RedSignal(x)$

We now know:

$ClearPath(Ambulance)$

Therefore:

GreenSignal(Ambulance)

and

¬RedSignal(Ambulance)

Derived Facts 2 and 3

Ambulance gets a green signal.

Ambulance does not have a red signal.

Step 3 — Apply Rule 3

Rule 3:

Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)

Choose:

x=CarA

and

y=Ambulance

We know:

Vehicle(CarA)
Behind(CarA,Ambulance)

and from Step 2:

GreenSignal(Ambulance)

All three conditions are satisfied.

Therefore:

Proceed(CarA)

Derived Fact 4

CarA is allowed to proceed.

Forward Chaining Table

Step	Rule Applied	Facts Satisfying Rule	New Fact
Initial	—	Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA,Ambulance)	Initial facts

Step	Rule Applied	Facts Satisfying Rule	New Fact
1	Rule 1	Vehicle(Ambulance) $\wedge$ EmergencyType(Ambulance)	ClearPath(Ambulance)
2	Rule 2	ClearPath(Ambulance)	GreenSignal(Ambulance)
3	Rule 2	ClearPath(Ambulance)	$\neg$ RedSignal(Ambulance)
4	Rule 3	Vehicle(CarA) $\wedge$ Behind(CarA,Ambulance) $\wedge$ GreenSignal(Ambulance)	Proceed(CarA)

(d) Final Knowledge Base After Inference

The system initially had:

Vehicle(Ambulance)
EmergencyType(Ambulance)
Vehicle(CarA)
Behind(CarA,Ambulance)
 After forward chaining, we additionally derive:

ClearPath(Ambulance)
GreenSignal(Ambulance)
 $\neg$ *RedSignal(Ambulance)*
Proceed(CarA)

Therefore, the final conclusions are:

ClearPath(Ambulance)

GreenSignal(Ambulance)

$\neg$ *RedSignal(Ambulance)*

Proceed(CarA)

Final Interpretation

The autonomous traffic management agent recognizes the Ambulance as an emergency vehicle and provides it with a clear path and green signal. Since CarA is behind the Ambulance and the Ambulance has a green signal, the system concludes that CarA can proceed as well.

Case Study 3 — Agricultural Expert Reasoning System

We are given the following propositional knowledge base:

P1: SoilDry $\rightarrow$ IrrigationNeeded

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Facts: SoilDry = True, CropWheat = True

Let us use the following symbols for simplicity:

Symbol	Meaning
S	Soil is dry
I	Irrigation is needed
W	Crop is wheat
D	Drip method is applied
R	Crop is at risk

(a) Convert P1, P2, P3 and Facts into CNF

What is CNF?

A formula is in Conjunctive Normal Form (CNF) when it is expressed as an AND of clauses, where each clause contains OR-connected literals.

For example:

$$(A \vee B) \wedge (C \vee D)$$

is in CNF.

The main conversion rules we need are:

Implication elimination

$$A \rightarrow B \equiv \neg A \vee B$$

Negation

$$\neg(\neg A) \equiv A$$

Distribution

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

P1: SoilDry $\rightarrow$ IrrigationNeeded

Using our symbols:

$$S \rightarrow I$$

Step 1 — Eliminate implication

$$S \rightarrow I \equiv \neg S \vee I$$

There are no further negations or conjunctions to distribute.

Therefore:

$$P1: \neg S \vee I$$

CNF Clause:

$$\{\neg S, I\}$$

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Symbolically:

$$(I \wedge W) \rightarrow D$$

Step 1 — Eliminate implication

$$\neg(I \wedge W) \vee D$$

Step 2 — Move $\neg$ inward

Using De Morgan's law:

$$\neg(I \wedge W) \equiv \neg I \vee \neg W$$

Therefore:

$$(\neg I \vee \neg W) \vee D$$

which simplifies to:

$$\neg I \vee \neg W \vee D$$

There is no conjunction remaining, so no distribution is required.

CNF Clause:

$$\{\neg I, \neg W, D\}$$

P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Symbolically:

$$(\neg D \wedge W) \rightarrow R$$

Step 1 — Eliminate implication

$$\neg(\neg D \wedge W) \vee R$$

Step 2 — Move $\neg$ inward

Using De Morgan's law:

$$\neg(\neg D \wedge W) \equiv \neg(\neg D) \vee \neg W$$

Since:

$$\neg(\neg D) = D$$

we get:

$$D \vee \neg W$$

Therefore:

$$(D \vee \neg W) \vee R$$

which simplifies to:

$$D \vee \neg W \vee R$$

CNF Clause:

$$\{D, \neg W, R\}$$

Facts in CNF

We have:

SoilDry=True

Therefore:

S

And:

CropWheat=True

Therefore:

W

A single propositional fact is already a valid CNF clause.

Complete CNF Knowledge Base

Therefore, the complete KB is:

$$(\neg S \vee I) \wedge (\neg I \vee \neg W \vee D) \wedge (D \vee \neg W \vee R) \wedge S \wedge W$$

Or as a set of clauses:

C
1

$$=\{\neg S, I\}$$

C
2

$$=\{\neg I, \neg W, D\}$$

C
3

$$=\{D, \neg W, R\}$$

C
4

$$=\{S\}$$

C
5

$$=\{W\}$$

(b) Resolution Refutation to Prove ApplyDripMethod

Our goal is:

$$D$$

To prove this using Resolution Refutation, we assume the opposite of the goal.

Step 1 — Negate the goal

Goal:

$$D$$

Negated goal:

$$\neg D$$

Add this to the CNF knowledge base.

So we now have:

$$\begin{array}{c} C \\ 6 \end{array}$$

$$=\{\neg D\}$$

Step 2 — Resolve the clauses

Our relevant clauses are:

$$\begin{array}{c} C \\ 1 \end{array}$$

$$\begin{array}{c} =\{\neg S, I\} \\ C \\ 4 \end{array}$$

$$=\{S\}$$

Resolve them on S:

$$(\neg S \vee I), S$$

Therefore:

$$I$$

Call this:

$$C7=\{I\}$$

So we have proved:

$$IrrigationNeeded$$

Step 3 — Resolve with CropWheat

We have:

$$C2=\{\neg I, \neg W, D\}$$

and:

$$C7=\{I\}$$

Resolve on I:

$$(\neg I \vee \neg W \vee D), I$$

Therefore:

$$\neg W \vee D$$

Call this:

$$C8=\{\neg W, D\}$$

Step 4 — Use the fact CropWheat

We have:

$$C5=\{W\}$$

Resolve:

$$(\neg W \vee D), W$$

Therefore:

$$D$$

Call this:

$$C9=\{D\}$$

So the KB proves:

ApplyDripMethod

Step 5 — Resolve with the Negated Goal

Remember that we added:

$$C6=\{\neg D\}$$

Now resolve:

$$D, \neg D$$

This produces the empty clause:

The empty clause represents a contradiction.

Therefore, our assumption:

$$\neg D$$

must be false.

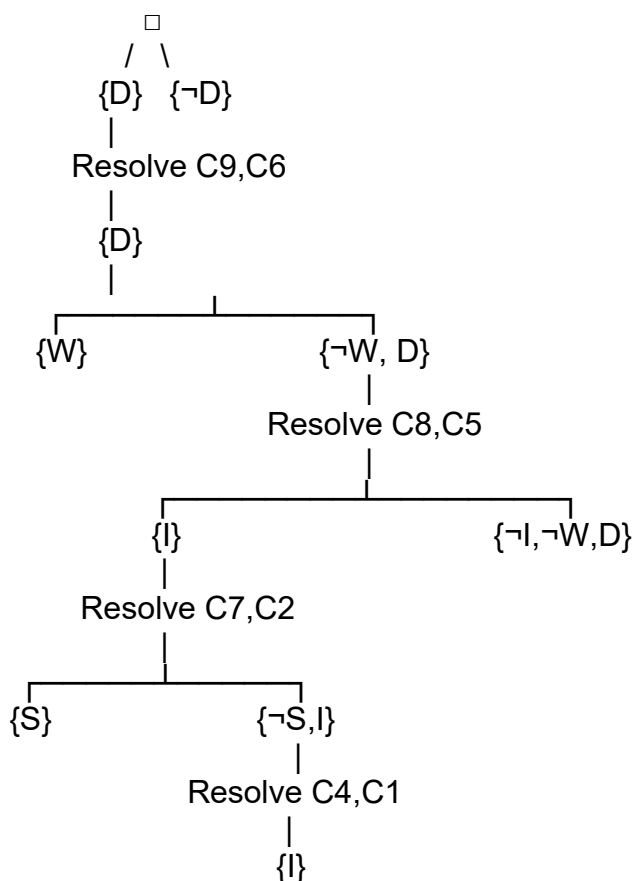
Hence:

D

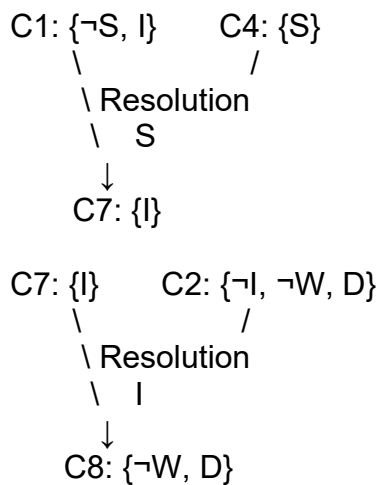
is logically entailed by the KB.

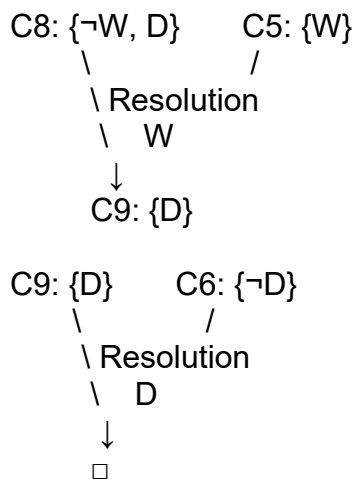
Resolution Proof Tree

The resolution proof can be shown as:



A cleaner bottom-up representation is:





Therefore:

KB ≠ ApplyDripMethod

(c) Remove the Fact SoilDry

Now suppose the fact:

SoilDry = True

is removed.

Therefore, we remove:

S

from the knowledge base.

The remaining KB is:

$(\neg S \vee I) \wedge (\neg I \vee \neg W \vee D) \wedge (D \vee \neg W \vee R) \wedge W$

Step 1 — Can we derive IrrigationNeeded?

Rule P1 is:

$S \rightarrow I$
or in CNF:

$\neg S \vee I$

But now we do not know *S*.

Therefore, we cannot establish:

I

So:

IrrigationNeeded cannot be derived

Step 2 — Can we derive ApplyDripMethod?

P2 requires:

$$I \wedge W$$

We still know:

$$W = \text{True}$$

but we no longer know:

$$I = \text{True}$$

Therefore, P2 cannot fire.

So:

ApplyDripMethod cannot be derived

Notice something important: "cannot be derived" does not mean "false."

The logical status of D is now unknown.

Step 3 — Does CropAtRisk follow?

P3 is:

$$\neg D \wedge W \rightarrow R$$

To derive R, we need:

$$\neg D$$

and:

$$W$$

We still have:

$$W = \text{True}$$

However, removing S does not give us:

$$\neg D$$

We only lost the ability to prove D.

Therefore:

D is unknown

does not imply:

$$\neg D$$

This is a very important principle of classical logic:

Not provable(D)

□

$=Proven(\neg D)$

Therefore, P3 cannot be applied.

CropAtRisk cannot be derived

Comparison Before and After Removing SoilDry

Proposition	With SoilDry	Without SoilDry
SoilDry (S)	True	Unknown
CropWheat (W)	True	True
IrrigationNeeded (I)	Derived	Not derivable
ApplyDripMethod (D)	Derived	Not derivable
$\neg$ ApplyDripMethod ($\neg D$)	Not given	Not derivable
CropAtRisk (R)	Not derivable	Not derivable

Important observation

Even before removing SoilDry, CropAtRisk does not follow.

Why?

Because we derive:

D

not:

$\neg D$

The rule for risk requires:

$\neg D \wedge W$

But we have:

$D \wedge W$

So the crop is not logically established as being at risk by P3.

Final Answer for (c)

After removing the fact SoilDry:

$S \in$
/
 KB

Therefore:

S

□

$\Rightarrow I$

so IrrigationNeeded cannot be derived.

Without I, Rule P2 cannot derive:

ApplyDripMethod

However, we also cannot conclude:

$\neg \text{ApplyDripMethod}$

because the absence of D is not evidence for $\neg D$.

Therefore P3:

$\neg D \wedge W \rightarrow R$

cannot fire.

Hence:

CropAtRisk does NOT follow.

The final logical status is:

I=Unknown, D=Unknown, R=Unknown

This demonstrates the difference between "false" and "not provable" in logical reasoning.

Case Study 4:

(a) Belief state and Modus Ponens derivations

Given percepts (facts)

From the problem:

- Stench[1,2]
- Breeze[1,1]
- Glitter[2,2]

Rules in the KB

1. Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
2. Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
3. Glitter[x,y] $\rightarrow$ Gold[x,y]
4. $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

Apply Modus Ponens

Using MP: if P and $P \rightarrow Q$, then infer Q .

1. From Glitter[2,2] and rule (3):
 - Gold[2,2]

2. From Stench[1,2] and rule (1):
 - WumpusAdjacent[1,2]
3. From Breeze[1,1] and rule (2):
 - PitAdjacent[1,1]

Now, using the usual Wumpus World adjacency (up, down, left, right) and assuming valid cells only (no 0 index):

- Cells adjacent to [1,2]:
 - [1,1], [1,3], [2,2] $\Rightarrow$ Possible Wumpus locations:
 - Wumpus[1,1] $\vee$ Wumpus[1,3] $\vee$ Wumpus[2,2]
- Cells adjacent to [1,1]:
 - [1,2], [2,1] $\Rightarrow$ Possible Pit locations:
 - Pit[1,2] $\vee$ Pit[2,1]

Belief state summary

The agent's belief state after inference:

- Percepts:
 - Stench[1,2], Breeze[1,1], Glitter[2,2]
- Derived facts (via Modus Ponens):
 - Gold[2,2]
 - WumpusAdjacent[1,2]
 - PitAdjacent[1,1]
- Possible locations (knowledge about world structure):
 - Wumpus $\in \{ [1,1], [1,3], [2,2] \}$
 - Pit $\in \{ [1,2], [2,1] \}$

No cell yet satisfies $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y]$, so rule (4) cannot mark any new cell as Safe.

(b) Forward chaining to find possible Wumpus and Pit locations

Forward chaining: start from known facts, repeatedly apply rules to derive new facts.

Step 1: Start with percepts

- Fact 1: Stench[1,2]
- Fact 2: Breeze[1,1]
- Fact 3: Glitter[2,2]

Step 2: Apply rules directly

- From Stench[1,2] and rule (1):
 - Infer: WumpusAdjacent[1,2]
- From Breeze[1,1] and rule (2):
 - Infer: PitAdjacent[1,1]
- From Glitter[2,2] and rule (3):
 - Infer: Gold[2,2]

Step 3: Expand adjacency into candidate cells

Assuming standard grid adjacency:

- WumpusAdjacent[1,2] $\Rightarrow$ Wumpus is in one of:

- $[1,1], [1,3], [2,2] \Rightarrow$ Possible Wumpus locations:
 - $Wumpus[1,1] \vee Wumpus[1,3] \vee Wumpus[2,2]$
- $PitAdjacent[1,1] \Rightarrow$ Pit is in one of:
 - $[1,2], [2,1] \Rightarrow$ Possible Pit locations:
 - $Pit[1,2] \vee Pit[2,1]$

Step 4: Check for Safe cells using rule (4)

Rule (4):

$$\neg Wumpus[x,y] \wedge \neg Pit[x,y] \rightarrow Safe[x,y]$$

We do not have any explicit $\neg Wumpus[x,y]$ or $\neg Pit[x,y]$ facts for $[1,1]$, $[1,2]$, or $[2,2]$. So no new $Safe[x,y]$ can be derived via forward chaining.

Conclusion of (b):

- Possible Wumpus locations: $[1,1], [1,3], [2,2]$
- Possible Pit locations: $[1,2], [2,1]$

(c) Is the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ safe?

We now evaluate each cell on the path using the knowledge from (a) and (b).

Cell $[1,1]$

- Percept: $Breeze[1,1] \Rightarrow PitAdjacent[1,1] \Rightarrow$ Pit at $[1,2]$ or $[2,1]$.
- From $Stench[1,2]$, Wumpus could be at $[1,1]$.
 - $Wumpus[1,1]$ is one of the candidates.

We cannot infer:

- $\neg Wumpus[1,1]$
- $\neg Pit[1,1]$ (we only know pit is adjacent, not that $[1,1]$ has no pit)

So rule (4) cannot give $Safe[1,1]$. The agent cannot logically guarantee that $[1,1]$ is safe.

Cell $[1,2]$

- Percept: $Stench[1,2] \Rightarrow WumpusAdjacent[1,2]$.
- From $PitAdjacent[1,1]$, $Pit[1,2]$ is a candidate location. So $[1,2]$ is:
- Adjacent to Wumpus, and
- Possibly containing a Pit.

Again, we do not have $\neg Wumpus[1,2]$ or $\neg Pit[1,2]$. So $Safe[1,2]$ cannot be derived. This cell is highly unsafe in the agent's belief.

Cell $[2,2]$

- Percept: $Glitter[2,2] \Rightarrow Gold[2,2]$.
- From $WumpusAdjacent[1,2]$, $Wumpus[2,2]$ is one of the possible locations. So $[2,2]$ may contain:
- Gold, but also possibly Wumpus.

Again, no $\neg Wumpus[2,2]$ or $\neg Pit[2,2]$. So $Safe[2,2]$ cannot be inferred.

Logical conclusion

Using rule (4):

$$\neg Wumpus[x, y] \wedge \neg Pit[x, y] \rightarrow Safe[x, y]$$

For cells [1,1], [1,2], [2,2]:

- We do not have both $\neg Wumpus$ and $\neg Pit$ for any of them.
- Therefore, the agent cannot prove any of these cells are safe.
- In fact:
 - [1,2] is a candidate Pit cell and adjacent to Wumpus.
 - [2,2] is a candidate Wumpus cell.
 - [1,1] is a candidate Wumpus cell (from adjacency).

Final verdict for (c)

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is *not logically guaranteed to be safe*. Based on the KB and derived knowledge:

- The agent has possible Wumpus and Pit on or near this path.
 - Since $Safe[x,y]$ cannot be inferred for any of these cells, the agent should not treat this path as safe to collect the gold.
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